

Reading the Lives of Stars

Colour-Magnitude Diagrams of the Globular Cluster M92 from the Iris Robotic Observatory

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Abstract. Iris is a self-operating robotic telescope that photographs deep-sky objects on its own each clear night. Beyond making pictures, it can turn its images into a measurement that astronomers have used for a century to understand stars: the colour-magnitude diagram, the observational form of the Hertzsprung–Russell (H–R) diagram. This note explains, for a reader with no background in astrophysics, what that diagram is, why the stars in it fall onto sharp lanes instead of a shapeless cloud, how those lanes record the life story of a star, and how the shape of the diagram lets us estimate the age of an entire star cluster. We use the observatory's own diagram of M92, a dense ball of hundreds of thousands of ancient stars.

1. What the observatory measured

M92 is a globular cluster: a gravitationally bound swarm of stars, all born together from the same cloud of gas, all sitting at essentially the same distance from us (about 27,000 light-years). That shared birth and distance is what makes it such a powerful natural laboratory, as we will see.

To build the diagram, Iris took many exposures of M92 through two colour filters — a blue one (B) and a red one (R) — stacked them to beat down noise, and measured the brightness of every star in each filter. It then calibrated those brightnesses against the European Space Agency's Gaia star catalogue, which has measured nearly two billion stars from space with exquisite precision. The result is one dot per star, placed on a graph by two numbers: its colour and its brightness.

2. The colour-magnitude (H–R) diagram

Every dot is a single star. Its position is set by two physical properties:

Horizontal axis — colour, which is really temperature. “Colour” here is a precise number: how much brighter a star looks in red light than in blue light ($R - B$). A hot star glows blue and sits to the left; a cool star glows red and sits to the right. So, counter-intuitively, the temperature runs hot-to-cold from left to right — a convention inherited from the original 1910s diagrams.

Vertical axis — brightness, which (because every cluster star is the same distance away) tracks the star's true power output, its luminosity. Brighter, more luminous stars are plotted toward the top.

In the figure, the dots are additionally tinted by their own colour — blue for hot, red for cool — so the temperature reading is reinforced visually. The vertical bar on the right is the key to that tinting.

3. The four panels of the M92 diagram

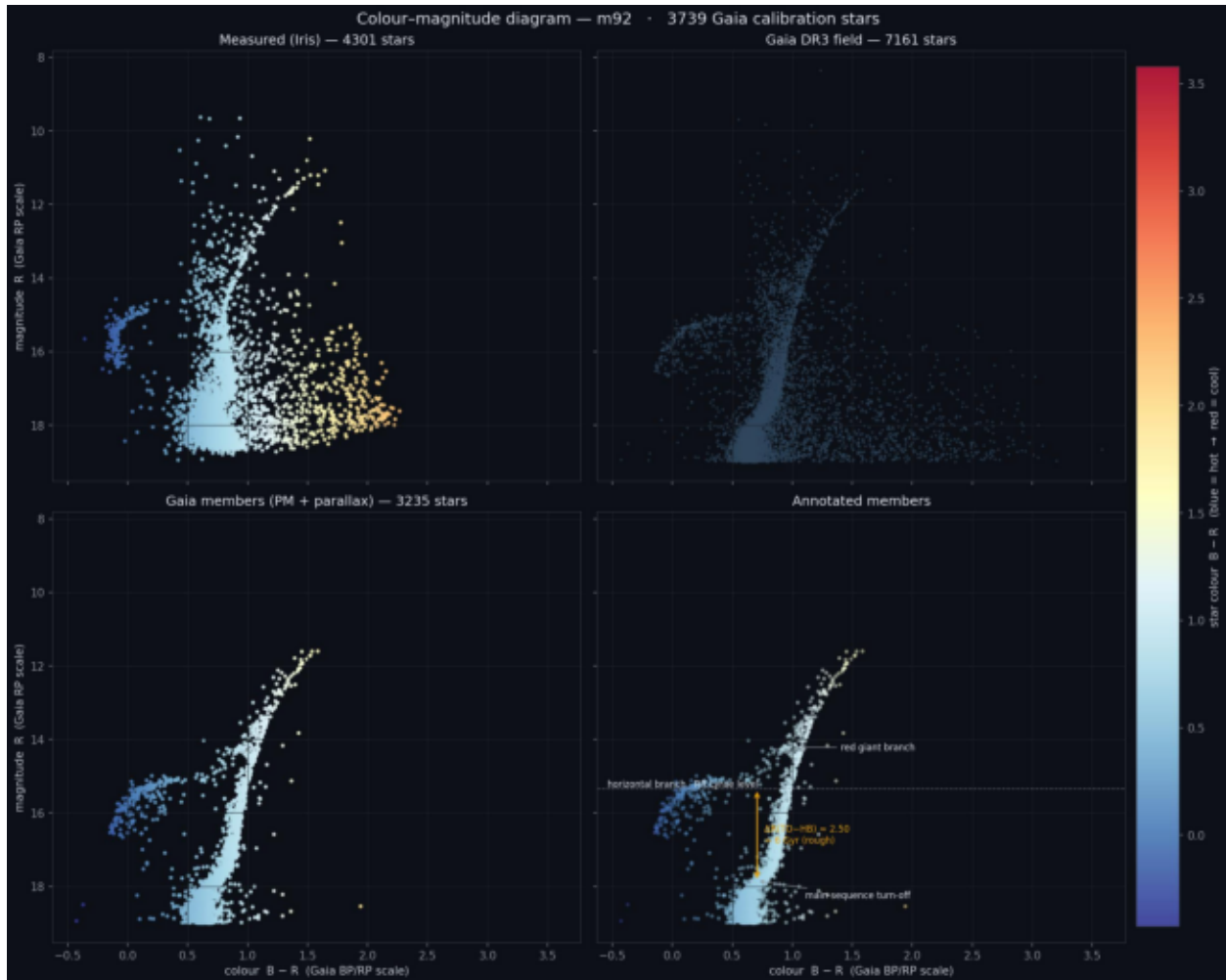


Figure 1. The colour-magnitude diagram of M92 produced by the Iris observatory. Four panels share the same axes: colour ($B-R$, hot to the left) horizontally and brightness (R , brighter at the top) vertically.

The figure presents the same cluster four ways, left to right, top to bottom:

(a) Measured (Iris). The diagram built entirely from the observatory's own B and R images — roughly 4,300 stars it detected and measured itself. The cluster's structure is already visible, but so is extra scatter: a haze of unrelated foreground stars from our own galaxy, and blurring where the crowded cluster core is hard to resolve. This is what the telescope genuinely saw.

(b) Gaia DR3 field. The same patch of sky as recorded by the Gaia space telescope — about

7,200 catalogued stars — shown in grey as an independent reference. It includes both true cluster members and the random foreground/background stars of the Milky Way, which is why it looks partly smeared. The measured panel should, and does, lie on top of this reference.

(c) Gaia members. Here the catalogue stars have been filtered down to the roughly 3,200 that are genuinely part of M92. Real members all drift across the sky together and lie at the same distance, so they share a common motion and a common parallax; foreground interlopers do not. Removing the interlopers strips away the haze and leaves the clean, true shape of the cluster — the textbook sequences emerge.

(d) Annotated members. The same clean diagram with its main features labelled automatically: the red giant branch, the main-sequence turn-off, and the horizontal branch (also called the RR Lyrae level). The orange caliper measures the vertical gap used to estimate the cluster's age, discussed in Section 6.

4. Why there is structure, not a cloud

The most striking thing about the members panel is what it is NOT: it is not a shapeless blob. The stars crowd onto a few narrow lanes, leaving most of the diagram empty. That is not an accident of sampling — it is physics.

A star is not free to have any brightness and temperature it likes. Once you fix a star's mass and its chemical make-up, the balance between gravity pulling it inward and the heat of nuclear fusion pushing outward pins it to one specific temperature and one specific luminosity — a single point on the diagram. A whole population of stars, born at the same time with a range of masses, therefore traces out a single continuous curve. The empty regions are simply combinations of brightness and colour that no stable, long-lived star can occupy.

The lanes also differ in how crowded they are, and that too is meaningful: stars spend about 90 percent of their lives in one stable phase and race through the others. The densely populated lane is where stars linger; the sparse regions are brief, fleeting stages of life caught in the act.

5. A star's journey across the diagram

The lanes are not random — they are the stages of a stellar life, and a star moves between them as it ages. Reading M92's members panel:

The main sequence is the long diagonal lane (mostly below this diagram's reach, picked up near the bottom). A star spends the vast majority of its life here, calmly fusing hydrogen into helium in its core. Where a star sits along it is set almost entirely by its mass: heavy stars are hot and bright (upper left), lightweight stars cool and faint (lower right).

The main-sequence turn-off is the hook where that lane bends upward and to the right. It marks the most massive stars that are just now running out of core hydrogen and leaving the

main sequence. As we will see, this hook is the cluster's clock.

The red giant branch is the steep lane climbing to the upper right. Having exhausted their core fuel, these stars swell enormously and cool at their surfaces, becoming luminous red giants — the fate that awaits our own Sun.

The horizontal branch is the spray of blue stars running leftward at a roughly constant brightness (the dashed line in the annotated panel, near magnitude 15). These older stars have ignited a new fuel — helium — in their cores and settled at a stable luminosity. Because M92 is very poor in heavy elements, its horizontal branch reaches unusually far to the hot, blue side. Stars caught in a narrow strip within this branch pulse in brightness over a matter of hours; these are the famous RR Lyrae variable stars, long used as cosmic distance markers.

So the diagram is a single snapshot that nonetheless shows a star's entire future laid out as a path: main sequence, to turn-off, up the red giant branch, over to the horizontal branch.

6. How the shape reveals the cluster's age

Here is the elegant part. Massive stars burn through their fuel far faster than lightweight ones — a heavyweight star can exhaust itself in a few hundred million years while a lightweight one lasts longer than the present age of the universe. In a cluster, where every star was born at the same moment, the heavy stars die first. So as the cluster ages, the main-sequence turn-off — the point of the most massive star still surviving on the main sequence — slides steadily downward to fainter, lower-mass stars.

The position of that turn-off is therefore a clock. A bright, high turn-off means a young cluster; a faint, low turn-off means an old one. To turn this into a number without needing to know the cluster's exact distance, astronomers measure the turn-off *RELATIVE* to the horizontal branch, whose brightness is nearly fixed and acts as a built-in yardstick. The larger that vertical gap, the older the cluster. That gap is exactly what the orange caliper in the annotated panel measures.

Applied to M92, this quick automatic measurement returns an age of several billion years — unmistakably an ancient system, which is the correct conclusion. (M92 is in truth about 13 billion years old, among the oldest objects known, formed not long after the Big Bang.) The method shown here captures the idea and the order of magnitude; pinning down the precise figure is the subject of the next section.

7. Limits of this measurement

The age read straight off the diagram should be taken as a rough, illustrative estimate, and it deliberately errs on the low side. Two honest caveats explain why. First, the turn-off is the single hardest feature to locate automatically, and our images only just reach it. Second, and more fundamentally, the standard age recipe is calibrated for a particular

scientific brightness scale, whereas Iris uses its own ordinary imaging filters; the turn-off-to-horizontal-branch gap is smaller on this scale, so the simple formula systematically under-reports the age.

Closing that gap — from ‘clearly ancient’ to a precise figure in billions of years — requires fitting detailed stellar-evolution models (isochrones) that fold in the cluster's distance, its dust dimming, and its chemical composition. That is a natural next step for the observatory's analysis pipeline.

8. Conclusion

From nothing but its own nightly images and a public star catalogue, the Iris observatory reconstructs a measurement that encodes how stars live, age, and die. The colour-magnitude diagram of M92 is not a cloud of points but a set of sharp lanes, because the laws of stellar physics permit stars only specific combinations of temperature and brightness, and because each lane is a distinct chapter of stellar life. The shape even keeps time: the slow downward creep of the main-sequence turn-off marks out the billions of years since the cluster's stars were born. A telescope that started the night to make a pretty picture ends it having measured the age of one of the oldest structures in the galaxy.